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Review

A Review of the Advances and Emerging Approaches in Hydrological Forecasting: From Traditional to AI-Powered Models

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Abstract

Hydrological forecasting has evolved rapidly in response to intensifying climate variability, increasing data availability, and advances in computational modeling. This review synthesizes developments from 2006 to 2025, examining four major forecasting domains: statistical approaches, physically based models, data-driven machine learning and deep learning techniques, and hybrid or emerging physics–AI frameworks. Recent literature shows a decisive shift toward integrated, data-rich systems that leverage remote sensing, IoT networks, and artificial intelligence to overcome limitations in traditional forecasting. While hybrid and physics-informed AI models achieve notable improvements in accuracy, lead time, and scalability, persistent challenges remain, especially regarding data scarcity, model interpretability, cross-basin generalization, climate non-stationarity, and operational computational demands. This review highlights these limitations and outlines future directions needed to strengthen hydrological forecasting as a tool for climate adaptation, early warning systems, and long-term water resource planning. By consolidating methodological advances and emerging gaps, the study provides insights into how hydrological forecasting can transition toward more resilient, transparent, and decision-oriented frameworks.

Keywords: hydrological forecasting; machine learning; deep learning; water resource management; hybrid modeling

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1. Introduction

Hydrological forecasting plays a vital role in managing water-related challenges by providing timely and informed predictions about the behavior of hydrological systems. It is primarily important in making better information on decision-making on flood management, drought mitigation, water system operations, water resources planning, and hydropower generation [1]. In recent years, the relevance of forecasting has increased further as water managers now require more reliable and longer lead-time predictions to cope with increasingly unpredictable hydro-climatic patterns [2,3].

Globally, it has become a critical component of climate adaptation strategies and disaster risk reduction initiatives. Due to the increased frequency and intensity of hydro-meteorological hazards, such as floods, droughts, and extreme rainfall events, countries

across all continents are investing in more advanced and accurate hydrological prediction systems [4–6]. In Europe, the European Flood Awareness System (EFAS) was developed to provide ensemble-based flood forecasts, aiding emergency responses and transnational coordination [7]. Meanwhile, North America utilizes the National Water Model, a hydrological modeling framework that simulates observed and forecast streamflow over the whole of the United States, and the North American Ensemble Forecast System (NAEFS), which combines weather forecast tools to provide high-resolution hydrologic forecasts that enhance flood risk management and water allocation planning [4,5]. In Asia, countries like China are pioneering seasonal streamflow forecasting using land surface and climate models, particularly in large basins such as the Yellow River [8]. It draws from a global hydrological forecasting system that can make use of real-time seasonal climate predictions from North American Multi-model Ensemble (NMME) climate models [9]. These examples illustrate how different regions tailor forecasting strategies to their hydrological and institutional contexts, highlighting the growing diversity of operational systems worldwide.

In contrast, many developing nations struggle to face significant challenges due to sparse hydrometeorological data, limited technical infrastructure, and insufficient institutional capacity [10]. These limitations hinder the reliability and accessibility of forecasts. In Africa, although challenged by data scarcity and infrastructural limitations, is taking actions through international collaborations. For example, CIMA Foundation's FloodPROOFS initiative in East Africa is operationalizing impact-based early warning systems [11,12]. Similarly, the SERVIR program, a joint initiative of NASA and USAID, provides Earth observation data and applications to support environmental decision-making in developing countries [13]. However, despite these efforts, gaps remain in sustained funding, long-term data collection, and the integration of forecasting tools into local decision-making frameworks, all of which are essential for building resilient hydrometeorological services [14–17].

There has been increasing pressure on water systems due to population growth, climate variability, urban expansion, and extreme weather events [18]. In addition, the frequency and intensity of hydrological disasters have surged over recent years, exacerbating vulnerabilities in developed and developing regions [19]. These compounding pressures have made conventional forecasting tools inadequate in certain contexts, pushing researchers to explore more flexible, data-rich, and computationally advanced approaches [20–22].

Hydrological forecasting offers a proactive means to address these issues. It is essential for flood control, drought mitigation, reservoir operations, irrigation scheduling, hydropower generation, and integrated watershed management, since it allows early warning and advanced planning possible [23–26]. For instance, real-time streamflow predictions can guide dam releases to prevent downstream flooding, while seasonal drought forecasts can optimize agricultural water allocation [11,12]. Forecast accuracy has therefore become a central component of risk-informed water management, particularly as extreme events become more frequent and more spatially variable [27,28].

As water-related risks continue to grow more complex under climate change, the demand for integrated forecasting frameworks that bridge physical processes, data analytics, and decision-making systems has become increasingly urgent [29]. Emerging approaches that combine traditional hydrological knowledge with machine learning, remote sensing, and high-performance computing are now viewed as necessary to capture non-linear watershed behavior and climate-driven uncertainties [30–32].

A conceptual overview of hydrological forecasting systems is presented in Figure 1, illustrating how hydro-meteorological inputs are transformed through statistical, physically

based, data-driven, and hybrid modeling approaches to produce operational forecasts for water resource management.

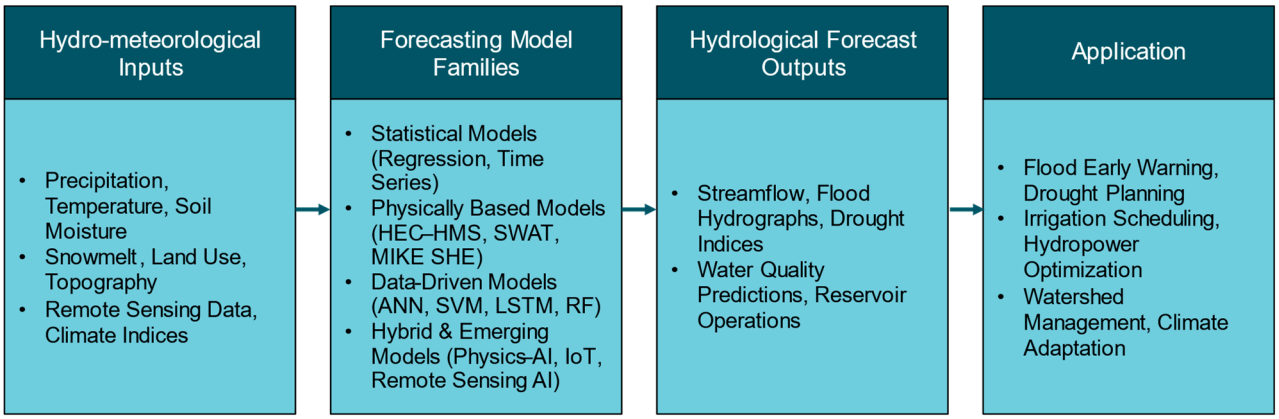


Figure 1. Conceptual framework of hydrological forecasting systems, showing how diverse hydro-meteorological inputs are processed through four major forecasting model families—statistical, physically based, data-driven, and hybrid/emerging approaches—to produce operational forecasts for water resources management.

This review focuses on recent trends and innovations in hydrological forecasting, covering both physically based and data-driven modeling approaches. It aims to synthesize developments in traditional forecasting systems, highlight advancements in AI, and machine learning applications, and compare the effectiveness and limitations of various forecasting techniques. Furthermore, it explores how these forecasting tools are integrated into water resource planning and disaster risk management, while also identifying emerging challenges, knowledge gaps, and future research directions in the field. By presenting an updated synthesis of forecasting developments from multiple methodological perspectives, this review contributes to a clearer understanding of how hydrological forecasting can evolve to support more resilient and adaptive water management strategies.

The remainder of the paper is organized as follows: Section 2 explains the literature selection process and thematic framework. Section 3 reviews major forecasting techniques and recent trends. Section 4 discusses applications in drought prediction, streamflow assessment, flood forecasting, water quality monitoring, and water scarcity analysis. Section 5 outlines emerging challenges and future directions, and Section 6 offers concluding insights.

2. Review Methodology

This review employs a meta-analysis approach to evaluate the current trends and innovations in hydrological forecasting, with a global perspective on emerging models, technologies, and applications. The objective is to synthesize insights across diverse geographic regions and hydrological conditions. The review draws from an extensive pool of scholarly references, including peer-reviewed journal articles, technical papers, institutional publications, and relevant online sources. To ensure systematic coverage, the review followed a structured screening and selection process with clearly defined search procedures, inclusion criteria, and thematic classification steps.

2.1. Literature Sourced and Selection Criteria

The literature search was designed to ensure a broad yet targeted selection of studies addressing hydrological forecasting methods and innovations. Peer-reviewed articles were primarily sourced from leading scientific databases and publication platforms such as Elsevier, MDPI, Springer, and other reputable academic publishers known for their

contributions to environmental science, hydrology, and water resources engineering. Additional searches were conducted through Scopus, Web of Science, Google Scholar, and IEEE Xplore to capture studies involving machine learning, deep learning, and hybrid modeling frameworks.

Selection criteria prioritized recent publications (ranging from 2006 to 2025) that demonstrated methodological novelty, practical application, or critical evaluations of forecasting models. References were gathered from institutional reports, conference proceedings, and international organizations specializing in hydrology and climate science. Studies were included based on their relevance to at least one forecasting domain and their contribution to the understanding of model development, performance evaluation, or operational implementation in varied hydrological contexts. Keyword combinations such as "hydrological forecasting", "streamflow prediction", "flood forecasting", "machine learning hydrology", "physically based models", and "hybrid hydrological models" were used, applying Boolean operators (AND/OR) to refine the search results.

Studies were excluded if they did not present original forecasting methodologies, lacked quantitative or reproducible model descriptions, or focused solely on socio-environmental discussions without hydrological modeling relevance. This ensured that only technically substantive works were incorporated into the synthesis.

To organize the review coherently, the selected studies were categorized under two overarching thematic lenses: (1) traditional forecasting tools, which include empirical statistical models, conceptual rainfall-runoff models, and physically based models; and (2) advanced forecasting tools, which comprise machine learning, deep learning approaches, and hybrid frameworks that integrate physical models with AI-based methods. Some studies also explored emerging trends such as remote sensing data assimilation, cloud-based modeling platforms, and Internet of Things (IoT) systems for real-time hydrological forecasting.

This structured screening ensured transparency and reproducibility, while the thematic grouping enabled a clearer comparison of methodological strengths, limitations, and application contexts.

This classification enabled a focused synthesis of modeling approaches, performance metrics, and context-specific applications across various hydrological domains.

2.2. Thematic Classification

To systematically analyze the reviewed literature, the references were grouped into four overlapping thematic categories. These groups summarize the focus of the research areas addressed across the studies and serve as a lens for examining trends, existing gaps, and potential directions in hydrological forecasting. It is important to note that the categories are not mutually exclusive, as many studies span multiple thematic areas. Thematic overlaps were preserved to reflect the increasing integration of physical models, machine learning, and hybrid approaches in recent forecasting research.

The first domain, traditional and statistical forecasting techniques, include the use of historical and statistical data to predict and forecast hydrological conditions. These techniques utilize functional linear models to predict streamflow, and a non-linear regression approach to model rainfall-runoff processes. These models remain important for baseline predictions and benchmarking machine learning advancements.

The second domain, physically based models, include the use of mathematical representation and physical laws to simulate natural hydrological processes. Models such as HEC-HMS, SWAT, and MIKE SHE proved to be effective in hydrological forecasting. These approaches are widely used in watershed-scale studies because of their ability to represent process interactions, despite requiring intensive calibration and high-quality datasets.

The third domain, it involves Machine Learning and Deep Learning models. These tools utilize historical datasets to predict hydrological responses without the need for explicit physical process descriptions. It covers ANNs, SVMs, DTs, RF, and many more models that could be used for hydrological forecasting tasks. Recent studies highlight their strong performance in nonlinear pattern detection, though issues of interpretability and overfitting remain prominent concerns.

Finally, a fourth domain involves the emergence of hybrid methods. A hybrid model combines two or more models for better prediction accuracy and to address the limitations of a single model. It includes the integration of Artificial Intelligence (AI) to offer notable developments in hydrological forecasting. Hybrid and physics-informed AI models are increasingly viewed as promising solutions for bridging data-driven flexibility with physical-process realism. Detailed quantitative distribution and geographical patterns are presented in Sections 3.5 and 4.6.

2.3. Temporal and Source Distribution

The publication years of the reviewed references were examined to evaluate the novelty and progression of research trends. Illustrated in Figure 2 are the references that were classified into six time-based categories. Notably, more than half of the sources were published between 2022 and 2025, emphasizing the focus of the review on contemporary developments and emerging innovations in hydrological forecasting. At the same time, the inclusion of earlier studies ensures continuity with foundational concepts and well-established modeling approaches. This distribution demonstrates a clear acceleration of research activity in the past five years, driven by advances in machine learning, remote sensing availability, and the increasing urgency of climate-linked hydrological extremes.

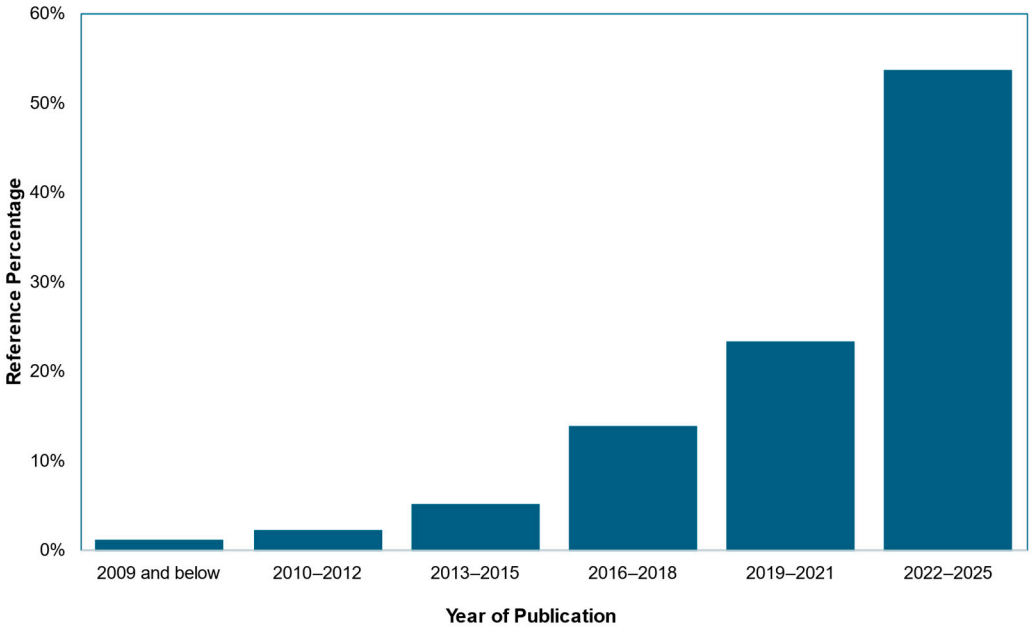


Figure 2. Reference distribution based on year of publication.

2.4. Reference Screening, Classification, and Meta-Analysis Procedure

To support the thematic synthesis presented in this review, a structured meta-analysis of all included references was conducted. The full reference list was screened and manually coded according to (1) forecasting technique used, (2) primary hydrological application domain, and (3) geographical region where the study was applied. Each paper was assigned to one or more technique categories—statistical models, physically based models, machine learning (ML), deep learning (DL), hybrid frameworks, and emerging

technology–enabled approaches. Application-based classification covered drought prediction, streamflow forecasting, flood risk assessment, water quality modeling, and water scarcity or allocation studies.

Geographical attribution was determined based on the stated study area or basin rather than the journal location, allowing for an accurate representation of global research distribution. All coded references were subsequently aggregated to compute technique frequencies, application frequencies, and regional distributions. Percentages were derived using total counts per category, and publication trends were assessed by grouping references into three-year intervals.

This procedure provides a consistent analytical basis for the quantitative summaries presented in Sections 3 and 4, ensuring that the synthesis reflects both methodological diversity and application-driven research trends across different geographical contexts.

3. Recent Trends in Hydrological Forecasting Techniques

Recent trends have progressed hydrological forecasting as researchers use various approaches to enhance hydrological forecasting continually [33,34]. Despite progress in methodologies, challenges still exist in the field of hydrological forecasting due to the changing and extreme hydrological events [34]. Following this, this section synthesizes current methods and explores the developments of hydrological forecasting, highlighting the key advancements in the field. Recent literature shows a clear shift toward integrated, data-rich, and adaptive forecasting systems that combine physical principles with artificial intelligence, reflecting the increasing need for resilience under climate uncertainty [35].

To avoid redundancy and improve conceptual clarity, this review deliberately separates methodological developments from application-specific discussions. Section 3 focuses on the technical foundations, modeling principles, and performance characteristics of hydrological forecasting approaches—including statistical, physically based, machine learning, deep learning, and hybrid models. Detailed discussions of how these methods are implemented in specific operational contexts such as flood forecasting, drought prediction, and water quality assessment are presented separately in Section 4. In the application-focused sections, methodological details are therefore not repeated; instead, cross-references to Section 3 are provided where appropriate.

3.1. Statistical Forecasting Techniques

In the field of hydrological forecasting, historical and statistical data are used to predict and forecast future hydrological conditions [36]. Masselot et al. (2016) [37] use functional linear models to predict streamflow based on precipitation curves instead of points. The paper highlights the potential of functional linear models, outperforming simple linear regression when it comes to the accuracy of volume forecasting. This innovative approach also suggested that FLM produced encouraging results as it forecasts the entire streamflow hydrograph, making it more effective for water resource management [38–40].

In addition, Safari et al. (2020) [41] used a non-linear regression approach to model rainfall-runoff. The Regression in the Reproducing Kernel Hilbert Space (RRKHS) was used to provide 1-,2-, and 3-day ahead forecasts using rainfall and streamflow data. This novel rainfall-runoff approach was compared with well-established models, which are the Multivariate adaptive regression splines (MARS) and Radial basis function artificial neural network (RBFNN), and proved its superiority in a more accurate regression analysis [42–44].

Moreover, a regression analysis of hydro-meteorological variables such as temperature, precipitation, humidity, and river flow was proposed by Baig et al. (2021) [45] using 30 years of observed data. Multiple regression was used to understand the relationship of the

variables to predict climate change. Promising results were shown in statistical parameters, and the model was able to effectively determine the relationship between variables.

Overall, statistical forecasting techniques remain highly interpretable and computationally efficient; however, their linear assumptions limit their ability to represent nonlinear hydrological responses. These constraints have prompted growing interest in more flexible machine learning and hybrid approaches capable of capturing complex dynamics, especially under extreme or rapidly changing climatic conditions [35,46,47]. Because of these limitations, hydrologists increasingly turn to physically based models, which explicitly simulate watershed processes using laws of physics rather than relying purely on historical data relationships.

3.2. Physically Based Models

In hydrological forecasting, physically based models use mathematical representation and physical laws to simulate the natural hydrological process [48]. The underlying principles often involved in this model are conservation of mass (water balance equations), Darcy's law, energy balance, and momentum and continuity equations to simulate and forecast hydrological events [49]. This model is commonly applied in different fields of hydrology, such as Flood forecasting and management, streamflow forecasting, rainfall-runoff, Drought forecasting, and Watershed management [48,50,51].

Consequently, models such as the Hydrologic Engineering Center's Hydrologic Modeling System (HEC—HMS), Soil and Water Assessment Tool (SWAT), and MIKE SHE have been supported by numerous studies recently to be effective in hydrological forecasting [52–54]. HEC-HMS 4.10 is a software developed for simulating rainfall-runoff systems in watersheds [55]. The applications that can be performed in this software include flood forecasting, watershed management, sediment transport, hydrological routing, and design of hydraulic structures. For instance, a study by Monjardin (2020) [56] used the HEC-HMS model to model rainfall-runoff and concluded that it has the capabilities to effectively replicate hydrological parameters. Moreover, SWAT integrates physical processes such as precipitation, interception, evapotranspiration, surface runoff, infiltration, percolation, and subsurface runoff [49]. It is a flexible tool that is commonly used to model runoff, groundwater flow, evapotranspiration, and non-point source pollution [57]. The study of Guug et al. (2020) [58] successfully incorporated SWAT to assess water availability using hydro-meteorological data in a catchment. Furthermore, MIKE SHE simulates hydrological events by integrating the interaction of surface and groundwater flow [59]. This model can be applied to applications such as watershed management, flood and drought forecasting, and ecosystem restoration [60].

In recent times, the physically based approach has been incorporated with other software and advanced models, such as AI, to improve its accuracy in hydrological forecasting and adapt to changing environments [61,62]. For instance, the integration of multiple advanced models, such as SWAT and Variable Infiltration Capacity (CIV), enables near-real-time forecasting of hydrological events or drought prediction to be specific [50]. Overall, physically based models offer more realistic simulations and flexibility in different scenarios, allowing for more accurate analysis of hydrological events. While this model established its effectiveness in hydrological forecasting, it often requires extensive data and resources, can be complex, and requires specialized expertise [63–65].

Physically based models are essential for capturing watershed-scale processes and ensuring physically realistic simulations. However, their heavy data requirements, calibration complexity, and computational intensity can limit their applicability in data-scarce basins [62,66]. These limitations have accelerated the rise of data-driven and machine learning approaches that offer improved flexibility and reduced dependence on detailed

process parameterization. As a result, researchers increasingly incorporate machine learning techniques that can model nonlinear hydrological behaviors without detailed physical formulations, especially where high-resolution data are available.

3.3. Data-Driven Models

The evolution of data-driven models has significantly transformed hydrological forecasting by leveraging statistical and machine learning (ML) techniques to model complex, nonlinear relationships fundamental in hydrological systems [67,68]. These models utilize historical datasets, such as precipitation, streamflow, temperature, and soil moisture, to predict hydrological responses without the need for explicit physical process descriptions. Furthermore, these are beneficial in scenarios where traditional physically based models face limitations due to data scarcity or limitations in computations [69].

3.3.1. Machine Learning (ML) Techniques

The emergence of ML introduced more refined algorithms that are capable of modeling complex patterns in hydrological data. Artificial Neural Networks (ANNs), Support Vector Machines (SVMs), Decision Trees (DTs), Random Forests (RF), and Gradient Boosting Machines (GBMs) have been applied to various hydrological forecasting tasks [70]. For instance, ref. [69] conducted a review that highlighted the efficacy of ML models in flood prediction, it emphasized their superior performance over traditional statistical methods in capturing nonlinear relationships.

Another notable example is the study [71] in the Andean watersheds of Southern Ecuador. They tackled the significant challenge of accurate streamflow forecasting in data-scarce and climatically variable regions. Farfán et al. (2020) [71] compared two physical models (WEAP and GR2M0 and two standalone ANN models, then developed a hybrid technique wherein the outputs from the physical and ANN models were used as inputs to a secondary ANN. This approach significantly improved forecast accuracy. The hybrid model raised the Nash-Sutcliffe Efficiency (NSE), outperforming all individual models. The result of the study [71] underscored the strength of ML in capturing hidden nonlinear patterns and enhancing prediction accuracy when combined with physically based insights.

In another context, Random Forest (RF) is a supervised machine-learning algorithm that constructs an ensemble of classification trees using bootstrap resampling of the training data and random subspace selection of predictor variables at each node split. This dual randomization strategy reduces overfitting and enhances model robustness. As demonstrated by Behnamian et al. (2017), RF's ensemble structure enables reliable assessment of variable importance; however, the stability of these importance measures depends on appropriate variable selection strategies and repeated model runs to mitigate randomness-induced variability [72]. Additionally, RF is a powerful machine learning algorithm that is widely used in hydrology for streamflow forecasting. It can capture complex, non-linear relationships between meteorological inputs and streamflow without requiring explicit physical modeling [73–75]. RF is especially useful for short-term forecasts due to its high predictive accuracy and ability to handle diverse watershed conditions, as shown by Pham et al., (2021) [76], who demonstrated its effectiveness across rainfall- and snowmelt-driven catchments.

Support Vector Machine (SVM) has been widely applied across various engineering disciplines for modeling and controlling complex systems [77]. Its strength lies in effectively capturing nonlinear relationships between input and output variables, which is particularly advantageous in hydrological modeling where such nonlinearities are common [78].

Machine learning methods excel at identifying nonlinear interactions in hydrological datasets, but their performance depends heavily on input data quality, appropriate feature

engineering, and robust model training. As hydrological systems often exhibit strong temporal dependencies, the emergence of deep learning has further improved performance in time-series forecasting tasks [79,80].

3.3.2. Deep Learning

Deep Learning (DL), a subset of ML, involves deep neural networks that are capable of learning hierarchical representations [81]. Long Short-Term Memory (LSTM) networks, a type of Recurrent Neural Network (RNN) [82], have shown remarkable success in modeling temporal dependencies in hydrological time series.

A study [83] applied Long Short-Term Memory (LSTM) networks to model storage effects in multiple catchments across the United States. The results showed that LSTM models outperformed traditional conceptual models in many basins, especially in capturing complex temporal patterns. Recent advances in remote sensing, cloud computing, and reanalysis datasets have further increased the feasibility and accuracy of data-driven hydrological forecasting [84–87].

Deep learning architectures such as LSTM, ConvLSTM, and Encoder–Decoder networks outperform traditional models in capturing spatial–temporal hydrological patterns. However, they often require large, high-resolution datasets and computational resources, making their application challenging in ungauged or poorly monitored basins [88–90].

3.3.3. Performance and Data Requirements

The performance of data-driven models in hydrological forecasting is highly dependent on the quality, quantity, and granularity of input data, as well as the structure and training of the chosen model [91,92].

In general, ML and DL models such as Random Forests, Support Vector Machines, LSTMs, and hybrid approaches have shown superior predictive accuracy compared to traditional conceptual models, particularly in modeling nonlinear processes and capturing temporal-spatial dependencies [93,94].

In terms of forecasting performance, models like LSTM and its variants consistently report high values of the Nash-Sutcliffe Efficiency (NSE), often exceeding 0.85 in daily or sub-daily streamflow predictions [83,95]. Hybrid deep learning models demonstrated a mean relative error of just 9.5% and achieved real-time flood mapping capabilities at speeds tens of thousands of times faster than traditional hydrodynamic simulations [95]. Meanwhile, physics-informed models [96] balanced accuracy with hydrological realism, where it showed significant improvements in terms of prediction while maintaining physical consistency, particularly in data-sparse basins.

Despite their effectiveness, these models have notable data requirements. At a minimum, most ML models require long-term, high-resolution time-series data on precipitation, temperature, and evapotranspiration [97–99]. For spatial models like ConvLSTM or Graph Neural Networks (GNNs), additional geospatial layers are needed, including land use, soil characteristics, digital elevation models (DEMs), and drainage networks [100–102].

Overall, data-driven approaches provide strong predictive capability but are highly dependent on the availability, resolution, and reliability of training data. These constraints motivate the integration of physics-based knowledge to ensure hydrological realism and avoid black-box behavior, especially in operational forecasting contexts [103,104]. To address the strengths and weaknesses of each model family, recent research has increasingly adopted hybrid frameworks that combine physical modeling with ML/DL capabilities.

3.4. Hybrid and Emerging Methods

In recent hydrological developments, hybrid models and emerging techniques have demonstrated their accuracy and adaptability in forecasting [105]. A hybrid model in

hydrological forecasting combines two or more models for better prediction accuracy and to address the limitations of a single model. Numerous hybrid models have been used in different studies: a study of Duan et al. (2023) [106] uses a hybrid physics–AI model, which combines a physically based model with deep learning to improve hydrological forecasting; Du and Pechlivanidis (2025) [107] integrate process-based hydrological models with statistical or machine learning to capture complex and non-linear hydrometeorological processes; and Nguyen et al. (2024) [108] present a hybrid hydrological model, which combines HEC-HMS and an Encoder–Decoder LSTM network, that resulted in an improved flood forecasting quality, especially at a longer forecasting time, and real-time flood forecasting.

Furthermore, integrating AI techniques has offered notable developments in hydrological forecasting, as it offers advanced modeling capabilities and non-linear hydrological processes [109]. It effectively integrates advanced technologies such as remote sensing technologies, in situ sensors, and imaging data sources. Researchers advocate the use of AI as it allows for improved decision-making and sustainable water management practices [110,111]. This emerging area, often termed Physics-Informed AI (PI-AI), is particularly adept at addressing highly complex, coupled physical systems that traditional models struggle with. For instance, PI-AI is now being utilized to model non-linear, geomechanical processes in the deep subsurface [112]. This includes complex, coupled fluid-solid mechanics, such as the modeling of seepage instability of karst collapse columns considering variable mass effect or the study of nonlinear evolution characteristics and seepage mechanics of fluids in broken rock mass, which are often analyzed using concepts from bifurcation theory [113]. Similarly, PI-AI can be used to simulate the influence of coupled processes like the energy evolution mechanism and fractal characteristics of rock failure on deep aquifer permeability, ensuring physically consistent predictions even in data-scarce, hydrogeologically complex basins.

In addition, hydrological models combined with remote sensing and IoT significantly impact hydrological forecasting by offering real-time data acquisition and simulation. For instance, Soo et al. (2024) [114] combined remote sensing and machine learning to enhance the simulation of streamflow in a river basin. The study tested ML models and concluded that the random forest model outperformed other models based on statistical parameters. The remote sensing data were acquired from rain gauge stations, and weather data from the NASA Global Land Data Assimilation System model. Similarly, an Internet of Things (IoT) based approach was studied by Barthwal (2023) [115] to acquire hydrological data for flash flood monitoring and forecasting. Furthermore, the IoT-based sensor enables real-time monitoring and acquisition of hydrological and meteorological data, which is then processed using an LSTM model to identify flood occurrences. The combination of both technologies has allowed for accurate flood prediction, which strengthens early warning systems when it comes to floods [116–119].

Hybrid and emerging approaches represent a major direction in modern hydrological forecasting. By merging physically based reasoning with the flexibility of AI, these models provide improved generalizability, reduced calibration burden, and enhanced real-time prediction ability. The integration of IoT sensors, remote sensing data, and cloud-based computation further accelerates the development of operational, high-resolution, and climate-resilient forecasting systems [120–122].

Structural Typologies of Hybrid Hydrological Forecasting Models

Recent hybrid hydrological forecasting studies increasingly adopt distinct architectural strategies to integrate physically based and data-driven models. While these approaches share the common goal of improving predictive performance, they differ substantially in their integration logic, information flow, and operational objectives. To im-

prove conceptual clarity, hybrid models can be broadly classified into serial, parallel, and ensemble-based architectures.

Serial hybrid models apply a sequential integration strategy, where the output of one model serves as the input to another. In hydrological forecasting, this typically involves using a physically based model to simulate intermediate hydrological states—such as runoff components, soil moisture, or baseflow—which are then provided as inputs to a machine learning or deep learning model for refinement. This architecture leverages physical process understanding while allowing data-driven models to correct systematic biases or capture residual nonlinearities. Serial hybrids are particularly effective in data-scarce basins, long-term forecasting, and applications requiring improved physical consistency.

Parallel hybrid models operate by running physically based and data-driven models simultaneously using the same meteorological inputs. The outputs of these independent models are subsequently combined, either through weighted averaging, error-based fusion, or secondary learning algorithms. This approach allows each model to contribute complementary strengths process realism from physical models and nonlinear pattern recognition from machine learning making parallel hybrids well suited for real-time forecasting and uncertainty-aware applications.

Ensemble-based hybrid models extend the parallel approach by integrating multiple models, often spanning statistical, physical, and machine learning techniques. Ensemble frameworks aggregate predictions using voting schemes, performance-weighted combinations, or meta-learning strategies. These models aim to reduce structural uncertainty and enhance robustness across varying hydrological conditions, particularly under non-stationary climate regimes and extreme-event forecasting. Structural typologies of hybrid hydrological forecasting models and their typical applications are presented in Table 1.

Table 1. Structural typologies of hybrid hydrological forecasting models and their typical applications.

Hybrid Type	Integration Logic	Typical Use Cases	Representative Applications
Serial Hybrid	Physical model outputs used as inputs to ML/DL models	Bias correction, long-term forecasting, data-scarce basins	SWAT → LSTM runoff refinement; HEC-HMS → ANN flood forecasting
Parallel Hybrid	Physical and ML models run independently; outputs fused	Real-time forecasting, uncertainty-aware prediction	WRF–HEC-HMS + RF streamflow fusion
Ensemble-Based Hybrid	Multiple models combined via weighting or meta-learning	Climate non-stationarity, extreme-event robustness	Multi-model flood early warning systems

This structural classification highlights that hybrid models are not a monolithic category but a family of architectures designed to address different forecasting challenges. Explicitly distinguishing their integration logic improves interpretability and supports informed framework selection in both research and operational settings.

3.5. Data Requirements and Dataset Specifications

One of the most significant challenges in the realm of hydrological forecasting is the intricate interplay of dataset availability, quality, and structure, which are crucial for the effective development, calibration, and validation of forecasting models. A thorough review of the existing literature reveals that the requirements for data can vary dramatically based on several factors, including the specific forecasting methodology employed, the scale of the modeling efforts, and the particular application domain in question [123].

For instance, traditional statistical models and physically based models are heavily reliant on well-established hydro-meteorological observations, which typically consist of long-term datasets that capture essential variables such as precipitation, temperature, and

streamflow. These models often necessitate a robust and consistent historical dataset to ensure accurate parameter estimation and model performance [124].

In contrast, data-driven and hybrid modeling approaches are becoming increasingly prominent in hydrological forecasting [125]. These methods leverage high-resolution datasets that are sourced from multiple platforms, including remote sensing technologies, ground-based observations, and other innovative data collection techniques. The reliance on spatio-temporal datasets allows for a more nuanced understanding of hydrological processes, enabling models to capture complex interactions and variations over time and space [126]. As a result, the demand for diverse, high-quality data has surged, highlighting the need for advancements in data acquisition, processing, and integration methodologies to support these modern forecasting techniques effectively.

Meteorological inputs are essential components that underpin all forecasting techniques in hydrological prediction, serving as the cornerstone for accurate and reliable assessments of water systems [127]. These inputs encompass a variety of atmospheric parameters, including but not limited to precipitation levels, which indicate the quantity and intensity of rainfall; air temperature metrics, such as minimum, maximum, and mean temperatures that influence evaporation and plant transpiration; relative humidity, which affects the moisture content in the atmosphere; wind speed, which can impact evaporation rates; solar radiation, a critical factor in determining evapotranspiration rates; and evapotranspiration itself, the process through which water is transferred from land to the atmosphere.

In addition to meteorological data, hydrological inputs play a vital role in the prediction models. These inputs typically involve observed streamflow measurements, which provide insights into the volume of water flowing in rivers and streams; water level data, which helps gauge the height of water bodies; soil moisture content, crucial for understanding water availability for plants and the potential for runoff; groundwater levels, which offer information about the saturation of aquifers; and reservoir storage capacity, which indicates the amount of water held in storage facilities [128]. These hydrological inputs can be utilized either as direct predictors in forecasting models or as calibration targets to enhance the accuracy of predictions.

Moreover, the integration of topographic and physiographic data is particularly significant for the development of physically based and spatially distributed models. This data includes digital elevation models (DEMs), which provide detailed representations of terrain and landforms; slope measurements that affect water flow and accumulation; drainage networks that illustrate how water moves across the landscape; soil properties that influence infiltration and retention; and land-use/land-cover (LULC) data, which reflects human activities and natural vegetation patterns [129]. Collectively, these datasets contribute to a comprehensive understanding of hydrological processes, enabling more precise modeling and forecasting of water resources in various environments.

Data-driven machine learning (ML) and deep learning (DL) models represent a significant evolution in predictive analytics, particularly in the fields of hydrology and environmental science [130]. Unlike traditional modeling approaches, which often rely on simpler datasets, ML and DL techniques necessitate extensive time series data with high temporal resolution. For effective flood forecasting and real-time applications, datasets with daily or even sub-daily records—such as hourly or 15 min intervals—are typically employed. This granularity allows for a more nuanced understanding of hydrological dynamics and the rapid changes associated with flood events.

Conversely, when it comes to drought prediction and long-term assessments of water availability, the focus shifts towards monthly or seasonal datasets. These longer time

scales are essential for capturing the gradual changes in climate and water resources that characterize drought conditions, thus enabling more accurate forecasting and planning.

In addition to temporal resolution, ML and DL models often leverage a variety of remotely sensed products. These include satellite-derived rainfall estimates, land surface temperatures, vegetation indices like the Normalized Difference Vegetation Index (NDVI), soil moisture content, and snow cover measurements [131]. Such data is particularly invaluable in regions where traditional ground-based observations are sparse or non-existent, enhancing the models' predictive capabilities in data-scarce environments.

Moreover, the integration of hybrid and physics-informed models marks a significant advancement in the modeling landscape. These models marry the data-driven strengths of ML and DL with the foundational principles of physics-based approaches. By incorporating outputs from conceptual or process-based models—such as simulated runoff, soil moisture, or baseflow—as supplementary inputs into ML or DL architectures, these hybrid models effectively reduce data dimensionality while maintaining a level of hydrological realism that is often lost in purely data-driven methods.

The evolution of forecasting systems is further bolstered by the incorporation of real-time data streams from an array of Internet of Things (IoT) sensors, weather radar systems, and reanalysis products [132]. This integration not only enhances the operational forecasting capabilities but also supports the development of early warning systems, allowing for timely interventions and improved disaster preparedness. As these technologies continue to advance, the potential for more accurate and responsive water resource management grows, paving the way for smarter and more resilient environmental strategies.

Table 2 provides a structured categorization of the commonly used datasets in hydrological forecasting, summarizing key input variables, output variables, typical temporal resolution, and primary data sources associated with each modeling approach. This synthesis highlights how increasing model complexity is closely linked to higher data demands and more diverse data sources.

Table 2. Typical dataset specifications used in hydrological forecasting models across different methodological approaches.

Forecasting Approach	Typical Input Parameters	Typical Output Variables	Temporal Resolution	Common Data Sources
Statistical Models	Precipitation, temperature, historical streamflow	Streamflow, runoff, water level	Daily to monthly	Ground gauges, hydrometeorological stations
Physically Based Models	Precipitation, temperature, evapotranspiration, DEM, soil type, land use, channel geometry	Streamflow, flood hydrographs, soil moisture, groundwater recharge	Hourly to daily	Gauge data, DEMs, soil maps, land-use maps
Machine Learning (ML)	Precipitation, temperature, humidity, antecedent streamflow, soil moisture, LULC	Streamflow, flood peaks, drought indices	Sub-daily to daily	Gauges, reanalysis datasets, satellite products
Deep Learning (DL)	Multi-variable time series (precipitation, temperature, streamflow), spatial grids, remote sensing features	Streamflow, flood extent, water level	Sub-daily to daily	Satellite rainfall, reanalysis, IoT sensors
Hybrid/Physics–AI Models	Meteorological inputs, physical model outputs (e.g., runoff, baseflow), remote sensing variables	Streamflow, flood forecasts, drought indicators	Real-time to daily	Gauges, hydrological models, satellite and IoT data

3.6. Dataset Length and Sample Size Consideration in Hydrological Forecasting

The length of hydrometeorological datasets plays a pivotal role in determining the robustness, stability, and generalizability of hydrological forecasting models. The duration of these datasets can significantly impact the accuracy and reliability of predictions [133]. Short datasets, typically characterized by records of less than 10 years, can lead to issues such as parameter underestimation or overfitting of models. This is particularly problematic in data-scarce regions, where available information may be insufficient to capture the complexities of hydrological processes. Conversely, excessively long datasets, those extending beyond 30 years, may introduce non-stationary climatic signals that complicate the calibration process, as they can reflect changes in climate patterns over time rather than stable relationships.

In reviewing the existing literature, it becomes apparent that the lengths of hydrometeorological datasets vary considerably, influenced by factors such as data availability, the specific objectives of forecasting, and the modeling approaches employed. To enhance clarity and facilitate comparability among studies, the datasets have been categorized into three distinct size classes based on the duration of their time-series records:

1. Small Datasets: These are defined as records shorter than 10 years. They are often utilized in regions where data is scarce, in pilot studies, or for real-time and event-based flood forecasting. Such datasets are increasingly prevalent in Internet of Things (IoT)-enabled systems, which are designed for rapid data collection and analysis. They are also commonly employed in flash flood studies and within newly established monitoring networks, where immediate data availability is crucial for timely decision-making.
2. Medium Datasets: This category encompasses records that range from 10 to 30 years. Medium datasets represent the most frequently encountered length in operational hydrological forecasting. They strike a balance between temporal coverage and data quality, making them particularly useful for a variety of modeling techniques, including machine learning, deep learning, and the calibration of physically based models. The moderate duration of these datasets allows for a more comprehensive understanding of hydrological trends while maintaining sufficient data integrity for reliable forecasting.
3. Large Datasets: Large datasets are those that exceed 30 years in length and are often sourced from long-term gauge networks, national hydrological services, or global reanalysis and satellite products. The extensive temporal coverage of these datasets makes them invaluable for applications sensitive to climatic variations. They are essential for long-term streamflow assessments, drought analysis, and cross-basin generalization studies, where understanding historical trends and variations is crucial for effective water resource management and planning.

In conclusion, the length of hydrometeorological datasets emerges as a critical factor that profoundly impacts the development, calibration, and application of hydrological forecasting models [134]. A nuanced understanding of how dataset size influences model performance is imperative for researchers and practitioners operating in this domain, as it can dramatically alter the accuracy and reliability of hydrological assessments, ultimately shaping the effectiveness of water management strategies.

An analysis of the reviewed studies reveals a distinct trend: machine learning and deep learning models are predominantly trained on medium to large datasets. This preference underscores their inherent demand for substantial amounts of data, as these models are particularly sensitive to variations in sample size [135]. Their performance often hinges on the availability of comprehensive datasets that allow them to capture complex patterns and relationships within the data, thereby enhancing predictive accuracy.

Conversely, physically based models and traditional statistical models, while capable of functioning with shorter datasets, demonstrate a marked improvement in stability and parameter reliability when they are calibrated using longer records. The extended datasets provide a richer context, allowing these models to better account for the intricate dynamics of hydrological processes, which are often influenced by a multitude of environmental factors.

Moreover, hybrid models and physics-informed approaches showcase a commendable resilience to the challenges posed by limited data lengths. By incorporating physically meaningful constraints and principles, these models can effectively mitigate the risks of overfitting that frequently plague small-sample contexts. This ability to leverage foundational scientific knowledge allows for a more robust modeling framework, even in scenarios where data availability is constrained.

Thus, the implications of dataset length extend far beyond mere statistical considerations; they resonate deeply within the realms of environmental science and resource management. As such, a thorough comprehension of these dynamics is essential for advancing the field and optimizing the use of hydrological forecasting models in real-world applications.

3.7. Summary and Synthesis

Table 3 provides an integrated summary of the major hydrological forecasting approaches reviewed in this study, highlighting their methodological foundations, applications, and key supporting literature. Statistical models remain the most traditional category, relying on historical data patterns and simple regression techniques. Their primary strength lies in interpretability and minimal data requirements; however, their linear assumptions and limited ability to represent nonlinear hydrological responses restrict their performance under extreme or non-stationary climatic conditions. Physically based models represent the opposite end of the methodological spectrum, simulating watershed processes through mass, momentum, and energy conservation laws. These models provide high hydrological realism and are widely used in rainfall–runoff modeling, flood prediction, and water-balance studies, yet they require extensive calibration, large datasets, and significant computational capacity.

Data-driven machine learning (ML) and deep learning (DL) approaches have gained prominence due to their capability to model nonlinear, multivariate, and temporally dependent hydrological behavior. ML methods such as Random Forests, Support Vector Machines, and Artificial Neural Networks demonstrate high predictive accuracy in diverse catchments, while DL architectures—especially LSTM, ConvLSTM, and encoder–decoder models—extend these capabilities by modeling long-sequence temporal dependencies and spatial heterogeneity. Despite their strong performance, these models require large, high-quality datasets and careful training to avoid black-box behavior. Hybrid forecasting techniques have emerged to address this gap by integrating physically based reasoning with ML/DL architectures, allowing models to retain hydrological interpretability while benefiting from advanced statistical learning. Emerging technologies—including IoT-enabled hydrological sensor networks, remote sensing–AI fusion approaches, and graph-based neural architectures—represent the next generation of forecasting systems focused on real-time monitoring and operational early warning.

Table 3. Summary of Recent Hydrological Forecasting Approaches.

Forecasting Technique	Description	Key References
Statistical Models	Use historical data and statistical relationships to predict hydrological variables	[36,37,41,45]
Physically Based Models	Simulate hydrological processes based on physical laws (e.g., mass/momentum conservation)	[48–51,55–60]
Machine Learning (ML)	Learn patterns from data without explicit process descriptions (e.g., RF, SVM, ANN)	[69–71,76–78,81,91,92,95–97,100–102]
Deep Learning (DL)	Uses deep neural networks (e.g., LSTM) to capture temporal-spatial dependencies	
Hybrid Models	Combine two or more models (e.g., physics–AI, HEC HMS + LSTM) to improve performance	[105–111,114,115]
Emerging Technologies	Integrate sensors, remote sensing, and AI for real-time, large-scale data assimilation	

Beyond this conceptual synthesis, a quantitative bibliometric assessment of forecasting-related studies further illustrates contemporary methodological preferences. Analysis of 148 forecasting-focused references revealed the following distribution: Machine Learning (24.32%), Deep Learning (19.59%), Physically Based Models (18.24%), Hybrid Models (16.22%), Statistical Models (12.16%), and Emerging Technologies (9.46%). This distribution reflects the accelerating shift toward data-driven hydrology, with ML and DL collectively comprising nearly 44% of the studies. Physically based models remain foundational, particularly in structured or well-monitored watersheds, while hybrid frameworks continue to expand as researchers seek to combine physical interpretability with high predictive performance. Statistical models, though increasingly surpassed by newer approaches, continue to serve as essential baselines, especially in data-scarce environments. Emerging technologies, although representing a smaller share, point toward fast-growing interest in real-time forecasting and sensor-rich systems. These proportions are illustrated in Figure 3, which shows the percentage distribution of hydrological forecasting techniques identified in the reviewed literature. Machine learning and deep learning approaches represent a significant portion, highlighting the growing adoption of data-driven methods in hydrological forecasting. Traditional statistical and physically based models still maintain a substantial share, reflecting their continued importance as foundational tools. Hybrid and physics-informed AI models, while less frequent, are emerging as promising approaches that combine the strengths of both data-driven and process-based methods. This distribution underscores the trend toward integrating advanced computational techniques with conventional modeling frameworks to enhance predictive accuracy and adaptability in diverse hydrological applications.

Geographical analysis of the reviewed papers shows that forecasting techniques vary widely across global regions, influenced by differences in data availability, climatic conditions, and technological infrastructure. East, Southeast, and South Asia represent the largest concentration of advanced ML, DL, and hybrid forecasting studies, particularly in China, India, the Philippines, Malaysia, Thailand, and South Korea. These regions increasingly integrate LSTM-based models, SWAT–ML hybrids, and remote sensing–AI fusion due to rapidly expanding earth observation datasets and dense hydro-meteorological monitoring networks. Europe also exhibits strong use of physically based and hybrid approaches, with countries such as the United Kingdom, Germany, Spain, and the Netherlands frequently applying MIKE SHE, conceptual hydrological models, and physics-informed AI frameworks. North and South America, particularly the United States, Canada, Brazil, and Ecuador, display a balanced mix of DL models, hybrid techniques, and physically

based simulations, often supported by large open-source datasets and long-term watershed monitoring programs.

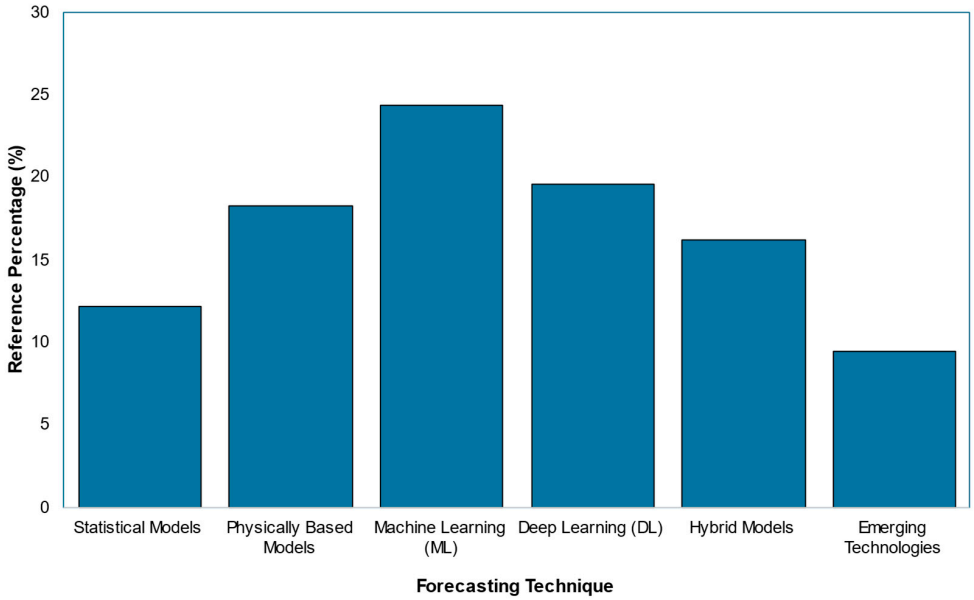


Figure 3. Percentage distribution of hydrological forecasting techniques identified from the reviewed literature.

In contrast, Africa shows increasing but more targeted applications, primarily focused on flood and drought forecasting using physically based and ML approaches in climatically vulnerable and data-limited basins. Studies from Ethiopia, Kenya, Malawi, and Zambia frequently employ HEC-HMS, SWAT, and RF-based early warning systems. The Middle East, including Saudi Arabia, Iran, and Turkey, shows strong use of physically based models combined with ML for arid-basin forecasting and drought analysis, reflecting the region's emphasis on water scarcity and long-term hydrological stress. These regional tendencies highlight that while ML and DL dominate globally, physically based and hybrid approaches remain essential in regions where hydrological realism and operational reliability are critical. The regional distribution patterns are summarized visually in Table 4, which presents a geographical checkbox comparison of forecasting techniques applied across major regions.

Table 4. Regional distribution of hydrological forecasting techniques based on the reviewed studies.

Region	Statistical	Physically Based	ML	DL	Hybrid	Emerging Tech
East Asia	✔	✔	✔	✔	✔	✔
South Asia	✔	✔	✔	✔	✔	—
Southeast Asia	—	✔	✔	✔	✔	—
Europe	✔	✔	✔	✔	✔	—
North America	✔	✔	✔	✔	✔	✔
South America	✔	✔	✔	✔	✔	—
Africa	✔	✔	✔	—	✔	—
Middle East	✔	✔	✔	—	✔	—

Note: A check mark (✔) indicates the presence of documented applications of each technique within a given region, while a dash (—) denotes either an absence of reported studies or insufficient information to confirm use.

Taken together, the methodological and geographical synthesis demonstrates that modern hydrological forecasting is characterized by a diverse and increasingly integrated set of approaches. The global literature reveals a clear movement toward flexible, data-rich, and hybrid forecasting frameworks capable of addressing intensifying hydrometeorological

extremes and climate uncertainty. As shown in Tables 3 and 4 and Figure 3, forecasting research continues to evolve toward multi-source data assimilation, physics-guided learning, and operational real-time prediction—signaling the maturation of a new generation of hydrological forecasting systems.

3.8. Unified Benchmark Comparison of Hydrological Forecasting Model Families

Despite the rapid expansion of hydrological forecasting approaches, direct comparison across model families remains challenging due to inconsistent evaluation criteria, heterogeneous datasets, and application-specific performance reporting. To address this limitation and improve the practical reference value of this review, a unified qualitative benchmarking framework is introduced to compare major hydrological forecasting model families across key operational dimensions.

Rather than focusing on case-specific numerical results which are often not directly comparable across studies this benchmark synthesizes findings from the reviewed literature using standardized criteria. These include predictive performance potential, computational demand, data requirements, interpretability, adaptability to non-stationary climate conditions, and suitability for real-time operations. This framework enables intuitive trade-off assessment among statistical, physically based, machine learning, deep learning, and hybrid models, supporting informed model selection for both research and operational contexts. Table 5 presents the unified benchmark comparison of hydrological forecasting model families across operational and methodological criteria.

Table 5. Unified benchmark comparison of hydrological forecasting model families across operational and methodological criteria.

Model Family	Predictive Capability	Data Requirements	Computational Demand	Interpretability	Climate and Scenario Adaptability	Real-Time Suitability	Typical Use Contexts
Statistical Models	Moderate (linear-dominated)	Low to moderate	Low	High	Low	High	Baseline forecasting, data-scarce regions, benchmarking
Physically Based Models	Moderate to high (process-consistent)	High (multi-source, long-term)	High	Very high	Moderate	Moderate	Watershed analysis, scenario simulation, regulatory planning
Machine Learning (ML)	High (nonlinear patterns)	Moderate to high	Moderate	Low to moderate	Low to moderate	High	Short-term forecasting, data-rich basins
Deep Learning (DL)	Very high (spatiotemporal learning)	High to very high	High	Low	Low	Moderate	Large-scale, high-resolution, real-time systems
Hybrid/Physics–AI Models	High to very high	Moderate to high	Moderate to high	Moderate to high	High	High	Climate-sensitive forecasting, operational decision support

4. Applications

While Section 3 reviewed recent advances in hydrological forecasting techniques from a methodological perspective, this section focuses on how these techniques are applied in practice across key hydrological domains. Rather than reiterating the technical foundations of statistical, physically based, and machine learning models, Section 4 emphasizes application-specific objectives, performance outcomes, and operational relevance in areas such as drought prediction, streamflow assessment, flood risk management, water qual-

ity forecasting, and water scarcity analysis. Methodological details are referenced from Section 3 where necessary to avoid duplication.

Hydrological forecasting provides essential support for water resources management, early warning systems, and climate resilience planning. Recent advancements in artificial intelligence (AI), machine learning (ML), remote sensing, and hybrid modeling have significantly expanded the practical applications of forecasting tools across multiple hydrological domains. These innovations have improved lead times, enhanced predictive accuracy, and strengthened decision-making in regions facing both data abundance and data scarcity. The following subsections present key applications and recent developments.

4.1. Drought Prediction and Management

Drought is one of the most complex and damaging hydrological hazards, with far-reaching environmental, economic, and societal impacts [136]. Its slow onset, long duration, and strong dependence on climatic variability make drought particularly difficult to forecast and manage [137]. These challenges are further exacerbated by climate change, which alters precipitation regimes, increases evapotranspiration, and disrupts historical drought patterns [138].

Recent applications of hydrological forecasting have therefore focused on improving drought prediction accuracy and lead time through the integration of advanced data-driven tools and multi-source datasets [139–143]. Rather than relying solely on traditional drought indices, recent studies increasingly apply artificial intelligence-based frameworks to enhance drought monitoring and forecasting capabilities. For example, Oyounalsoud et al. (2024) [136] evaluated multiple AI-based approaches—including Decision Tree (DT), Generalized Linear Model (GLM), Support Vector Machine (SVM), Artificial Neural Network (ANN), deep learning, and Random Forest (RF)—for drought prediction using climate and soil moisture data. Their results demonstrated that AI-based drought indices can outperform conventional indices in terms of reliability, robustness, and predictive accuracy, with the GLM-based index showing superior performance among the tested models.

Remote sensing–driven drought applications have also gained prominence, particularly in regions with sparse ground-based observations. Li et al. (2023) [144] integrated multisource satellite data with machine learning techniques to improve drought monitoring in Southwest China. By combining land surface temperature, precipitation, and vegetation indices derived from satellite and meteorological datasets, the study demonstrated that RF and eXtreme Gradient Boosting (XGBoost) models can effectively reconstruct missing data and provide consistent drought predictions. Such applications highlight the operational value of satellite-based drought forecasting systems, especially in data-scarce environments [143,145,146].

Overall, recent drought forecasting applications emphasize proactive management through early warning systems, long-term monitoring, and spatially distributed assessments. While data-driven approaches significantly improve predictive performance, their sensitivity to climate variability underscores the importance of hybrid frameworks that integrate physical drought indicators with machine learning tools to ensure robustness under non-stationary conditions [147–151].

4.2. Streamflow Assessment

Accurate streamflow forecasting is fundamental to flood risk mitigation, drought preparedness, reservoir operation, and integrated water resource management [152]. In recent years, hydrological forecasting applications have increasingly shifted toward data-driven and hybrid systems that improve both short-term operational forecasting and long-term planning [110,153–155]

Long-term streamflow forecasting remains particularly challenging due to climatic variability and hydrological nonlinearity. To address this, Balthazar et al. (2024) modeling) [156] applied supervised machine learning techniques under drought scenarios to enable long-range streamflow prediction. Using ridge regression and gradient boosting with hydroclimatic inputs, the study demonstrated that data-driven models can provide valuable decision-support information for long-term water resource management. Similarly, Liu et al. (2024) [157] employed a Relevance Vector Machine (RVM) model for long-term streamflow forecasting and showed that it achieved higher predictive accuracy than Support Vector Machines, highlighting its robustness for extended forecasting horizons.

In contrast, real-time and short-term streamflow applications prioritize rapid response and high temporal resolution. Zhang et al. (2024) [153] developed a hybrid machine learning framework that integrates networks of real-time monitoring stations for daily streamflow forecasting. The Spatio-Temporal Attention Gated Recurrent Unit (STA-GRU) model demonstrated superior performance, illustrating the importance of attention mechanisms and sensor networks in operational forecasting contexts.

Collectively, these applications show that modern streamflow forecasting systems benefit from flexible, data-rich modeling frameworks capable of addressing both long-term planning needs and real-time operational demands. Their integration with remote sensing, real-time monitoring, and decision-support systems significantly enhances basin-scale water management under variable hydrological conditions [158,159].

4.3. Flood Risk Management and Early Warning Systems

Flood forecasting is one of the most critical applications of hydrological prediction, supporting early warning systems, emergency response, and infrastructure protection worldwide [160,161]. Recent applications increasingly combine physically based models, data-driven approaches, and hybrid frameworks to improve forecast accuracy, lead time, and operational efficiency.

Physically based models remain central to flood forecasting applications due to their ability to represent watershed processes and simulate flood hydrographs. The Hydrologic Modeling System (HEC-HMS) continues to be widely applied in operational flood warning systems [162]. For example, Huang et al. (2024) [100] applied HEC-HMS in the XiHanShui River Basin, China, achieving strong agreement between simulated and observed flood events and enabling the identification of rainfall thresholds for real-time warnings. Similarly, González-Cao et al. (2019) [163] demonstrated the effectiveness of HEC-HMS in an automated early warning system for the Miño River basin, where simulated flows triggered subsequent hydraulic modeling to refine flood risk assessments. Integration with meteorological models further enhances flood forecasting performance, as shown by Goodarzi et al. (2024) [162], who coupled the Weather Research and Forecasting (WRF) model with HEC-HMS to improve flood prediction accuracy in a complex Iranian catchment. In urban and peri-urban environments, Koltsida et al. (2023) [164] confirmed the reliability of the SWAT model for runoff simulation and flood assessment.

Data-driven flood forecasting applications increasingly complement these physically based systems by offering rapid computation and adaptability across spatial scales [151,165]. Zhang et al. (2025) [166] applied a CNN–LSTM–Attention model for daily streamflow forecasting in cold-region basins, achieving high predictive accuracy and improved interpretability through SHAP analysis. Comparative studies by Farfán-Durán and Cea (2024) [154] further demonstrated the effectiveness of deep learning models for short lead-time flood forecasting, particularly when future rainfall information is included.

In highly flood-prone regions, Zhang et al. (2025) [166] showed that machine learning and deep learning models can provide reliable rainfall prediction and flood risk assess-

ment using long-term climatic datasets. Tang et al. (2023) [167] further illustrated the value of hybrid approaches by integrating machine learning-based event classification with dynamic parameter adjustment in conceptual models, enhancing real-time flood forecasting adaptability.

Overall, flood forecasting applications benefit most from integrated systems that combine physical realism, data-driven accuracy, and operational speed, enabling timely and actionable early warning services.

4.4. Water Quality

Water quality forecasting is an increasingly important application of hydrological modeling, as water quality dynamics influence ecosystem health, public safety, and sustainable water resource management [168]. Traditional water quality monitoring approaches are often labor-intensive and limited in spatial and temporal coverage, prompting the adoption of advanced forecasting tools [169,170].

Recent applications demonstrate that machine learning-based water quality forecasting can significantly improve prediction accuracy and operational efficiency [171,172]. Abbas et al. (2024) [173] applied multiple ML models—including XGBoost, RF, Gradient Boosting, and SVM—to predict the Water Quality Index (WQI), achieving high predictive accuracy and demonstrating the practicality of AI-based water quality assessment. Similar findings were reported by Sidek et al. (2024) [174] where ensemble ML models accurately predicted WQI across river basin samples. To enhance model transparency, Nallakaruppan et al. (2024) [175] incorporated Explainable Artificial Intelligence (XAI) techniques, improving interpretability and trust in ML-based water quality predictions.

Real-time and spatially distributed water quality forecasting has further advanced through the integration of remote sensing and hybrid modeling. Xiong et al. (2025) [176] combined in situ monitoring, process-based modeling, and machine learning to provide near real-time water quality forecasts at the bay scale, demonstrating applicability to rivers, reservoirs, and coastal systems. In data-scarce and heterogeneous environments, Zheng et al. (2025) [177] developed a deep learning framework capable of cross-basin water quality prediction, achieving high predictive performance across numerous monitoring sites.

Earlier applications, such as the work by Chang et al. (2015) [178], demonstrated the effectiveness of data-driven models in capturing hydrological influences on water quality dynamics, particularly during extreme events. Collectively, these studies highlight a clear trend toward automated, interpretable, and real-time water quality forecasting systems that support proactive environmental management.

4.5. Water Scarcity and Allocation

Water scarcity and allocation challenges have intensified due to climate variability, population growth, and competing sectoral demands. Hydrological forecasting applications increasingly support sustainable water management by informing infrastructure planning, allocation strategies, and long-term adaptation measures [179].

Doost and Yaseen (2024) [180] addressed water scarcity in arid and semi-arid regions by integrating GIS-based multi-criteria analysis with machine learning and physically based runoff estimation to optimize dam site selection. Their application demonstrated how spatial decision-support systems can enhance water harvesting and reduce vulnerability to drought. In urban and climate-sensitive regions, Estrada et al. (2024) [181] applied a hybrid ANFIS-based rainfall forecasting framework to improve precipitation prediction, supporting more responsive water allocation planning.

Accurate precipitation and long-term hydrological forecasting remain central to water scarcity management [182,183]. Recent applications highlight the growing role of

AI-enhanced weather and rainfall forecasting systems [184,185]. Advances in numerical weather prediction, remote sensing, and high-performance computing have enabled improved precipitation forecasts with longer lead times [185]. Bodnar et al. (2025) [186] introduced the Aurora AI model, demonstrating how large-scale AI systems can enhance high-resolution weather forecasting. Similarly, Yadav et al. (2025) [187] showed that ML and DL approaches can improve forecasting in remote, data-scarce regions, directly benefiting water resource planning.

Long-term rainfall forecasting applications using AI-based and statistical recurrent models have also proven valuable in arid climates. Alsumaiei (2025) [188] demonstrated that Neural Auto-regressive models can effectively replicate rainfall dynamics and support operational water management planning under climate uncertainty.

Overall, water scarcity and allocation applications increasingly rely on integrated forecasting frameworks that combine climate-informed prediction, spatial analysis, and AI-driven decision support to enhance resilience under growing hydrological stress [180–190].

4.6. Summary and Synthesis

Table 6 summarizes the dominant models applied across major hydrological forecasting domains, including drought prediction, streamflow assessment, flood forecasting, water quality evaluation, and water scarcity analysis. Across these applications, the reviewed studies demonstrate a consistent shift toward integrated, multi-source, and AI-enhanced modeling frameworks. Physically based models remain essential where process representation and hydrological realism are required, while machine learning (ML) and deep learning (DL) approaches deliver strong performance in capturing nonlinear relationships, leveraging high-dimensional datasets, and enabling real-time forecasting. Hybrid systems—which combine physical hydrological principles with ML/DL architectures—consistently emerge as the most adaptive and accurate across diverse basin conditions.

Table 6. Summary of findings of the models applied to several issues.

Title/Category	Models Used	Key Findings
Drought Prediction and Management	Hybrid Models (AI Models and ML and Remote Sensing)	Reliable, accurate robustness of data High consistency and advantageous in areas with sparse meteorological data
Streamflow Assessment	AI, Deep Learning, remote sensing, and Hybrid Models	Advantageous in long-term streamflow forecasting Yielding accurate and real-time forecasting
Flood Risk and Early Warning System	Physically based Models (HEC-HMS, SWAT) Data-driven Models (ML and DL) Hybrid Models (e.g., CNN-LSTM)	Real-time flood risk monitoring system Enhanced flood forecasting performance
Water Quality	Statistical Techniques Data-driven Model (ML) Hybrid Models (RF and Gradient Boosting Regression)	Accurate and precise quality assessment
Water Scarcity and Allocation	Data-driven Models (ML and DL) Hybrid Models (e.g., ANFIS, Aurora AI Model)	Sustainable solutions for water harvesting and allocation Accurate rainfall forecasting for managing limited water resources

Beyond methodological synthesis, the distribution of forecasting applications reveals clear research priorities. A quantitative analysis of the reviewed studies shows that streamflow forecasting constitutes the largest share of recent application-focused research (31.93%), reflecting its central role in operational water management and flood risk mitigation. Flood forecasting and early warning systems follow closely (27.73%), with strong representation

of physically based, ML, and hybrid models. Drought prediction and management (18.49%) and water quality forecasting (11.76%) show growing methodological diversification, supported by remote sensing, IoT-based monitoring, and DL-driven time-series modeling. Water scarcity and allocation studies (10.08%) increasingly incorporate ML-based precipitation prediction and hybrid optimization frameworks to support adaptive resource allocation under climatic stress. Figure 4 illustrates the percentage distribution of hydrological forecasting applications across the reviewed studies. Flood monitoring and early warning emerge as the most frequently addressed application, reflecting the critical need for timely hazard mitigation. Drought prediction and water allocation management also constitute significant portions, highlighting the role of forecasting in supporting sustainable water resources planning. Less common applications, such as water quality monitoring and integrated watershed management, indicate emerging areas where forecasting tools are increasingly being explored. Overall, this distribution emphasizes the diverse and growing utility of hydrological forecasting in addressing both hazard preparedness and water resource sustainability.

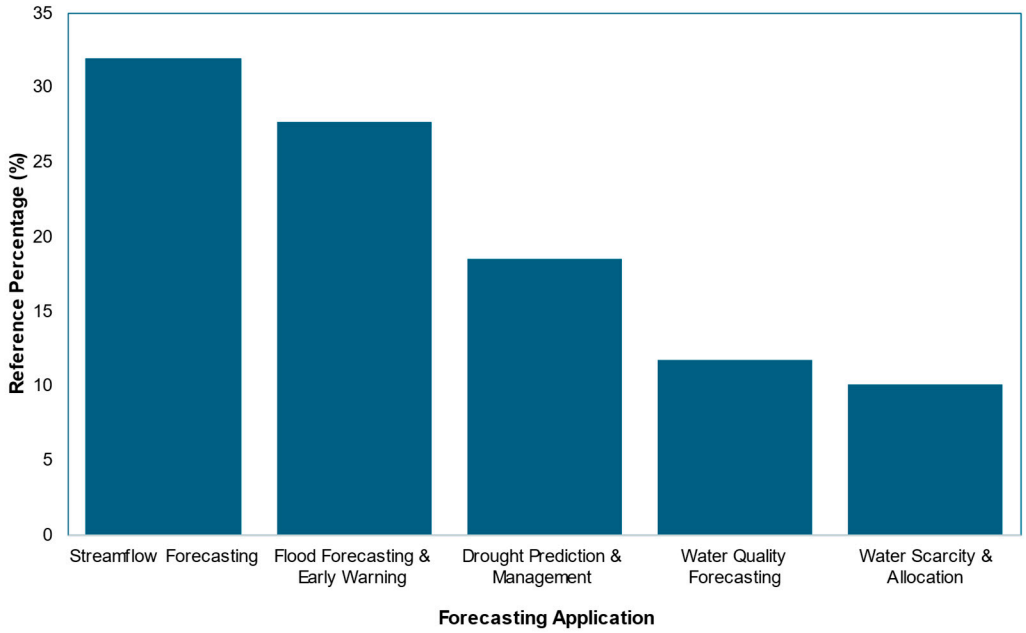


Figure 4. Percentage distribution of hydrological forecasting applications across the reviewed studies.

Geographical analysis further reveals that forecasting applications are strongly shaped by regional climatic conditions, hydrological challenges, and data availability. As shown in Table 7, Asia (East, South, and Southeast Asia) demonstrates strong emphasis on streamflow, flood, and drought forecasting—consistent with monsoon-driven hydrology and high exposure to extreme rainfall events. Europe and North America show a more balanced distribution, including significant contributions to water quality forecasting supported by dense monitoring networks and long-term environmental datasets. South America focuses primarily on streamflow and flood forecasting in tropical and mountainous catchments, while Africa and the Middle East prioritize drought prediction and water scarcity modeling due to chronic aridity, rainfall variability, and limited hydrological infrastructure.

Across all regions, hybrid and AI-based models consistently outperform traditional approaches, offering improved robustness, higher temporal–spatial fidelity, and enhanced operational value. Streamflow assessment benefits from attention-based and encoder–decoder DL architectures; flood forecasting achieves improved lead times through combined HEC–HMS, ML, and hybrid neural networks; water quality forecasting leverages

ML for spatiotemporal pollutant mapping; and water scarcity applications utilize ANFIS and ML-enhanced precipitation models for resource allocation under uncertainty.

Table 7. Geographical distribution of hydrological forecasting applications.

Region	Streamflow	Flood	Drought	Water Quality	Water Scarcity
East Asia	✔	✔	✔	✔	✔
South Asia	✔	✔	✔	—	✔
Southeast Asia	✔	✔	—	—	✔
Europe	✔	✔	✔	✔	—
North America	✔	✔	✔	✔	—
South America	✔	✔	—	✔	—
Africa	✔	✔	✔	—	✔
Middle East	✔	—	✔	—	✔

Note: A check mark (✔) indicates that the reviewed literature reports studies in the region for the corresponding application area, while a dash (—) denotes no identified studies in the dataset.

Taken together, the findings in Tables 6 and 7, and Figure 4 reinforce the conclusion that modern hydrological forecasting increasingly relies on integrated, AI-augmented, and multi-source modeling approaches. These systems not only enhance predictive accuracy but also provide stronger interpretability, greater adaptability, and improved operational relevance under conditions of climate variability, data limitations, and watershed complexity. As forecasting demands intensify globally, the continued advancement of hybrid, physics-informed, and sensor-integrated forecasting techniques will be critical for building resilient and sustainable water management systems worldwide.

5. Challenges and Solution-Oriented Future Directions in Hydrological Forecasting

Recent advances in hydrological forecasting have been driven by statistical, physically based, and data-driven approaches, with increasing integration of AI, remote sensing, and IoT technologies. Figure 5 shows the conceptual overview of how recent and emerging hydrological forecasting is applied to water management, highlighting how it improves early warning systems and real-time monitoring, makes informed decisions, and improves existing policies. However, despite the rapid advancement in hydrological forecasting, several challenges persist that limit its practical and operational applications [69,83,191].

Figure 5 highlights that hydrological forecasting now operates within a larger decision-support framework rather than as an isolated prediction tool. Its components—statistical, physically based, data-driven, and hybrid approaches—feed into diverse applications such as drought prediction, flood risk management, water quality assessment, and water allocation. This transition toward integrated, application-oriented forecasting further emphasizes the importance of addressing existing limitations to ensure forecasts translate into meaningful action in water resource planning and climate resilience strategies.

5.1. Data Limitations and Infrastructure Constraints

One of the most critical issues is the availability of forecasting models. Physically based models such as HEC-HMS, SWAT, and MIKE SHE require extensive meteorological and hydrological datasets to accurately simulate progress [48]. These data requirements pose a challenge, especially in regions with sparse observational records or limited infrastructure.

For data-driven models, the quality, quantity, and granularity of historical data remain essential factors. Models such as RF, SVM, and LSTM networks heavily rely on long-term, high-resolution time-series data for parameters. The absence of such data can reduce model accuracy and reliability [83,178].

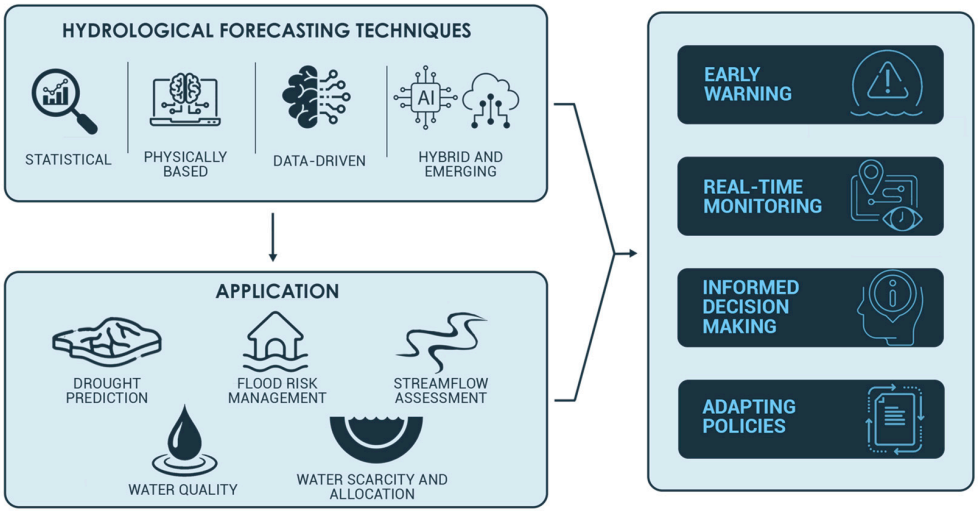


Figure 5. Conceptual Overview of the Role of Recent and Emerging Hydrological Forecasting in Water Management.

In many developing regions, hydrological data are further affected by fragmented monitoring networks, inconsistent calibration of instruments, missing data records, and the absence of automated real-time monitoring [192–194]. Limited access to satellite-based rainfall products or high-resolution reanalysis data adds another layer of complexity. As a result, many basins remain "data-scarce", restricting the adoption of advanced forecasting models. Enhancing observational infrastructure and integrating diverse data sources—ground-based sensors, remote sensing, and IoT networks—remains a critical future priority [62,195,196].

5.2. Limited Model Transparency and Trustworthiness

Additionally, although Explainable Artificial Intelligence (XAI) techniques have been introduced in the field of water quality forecasting to improve model transparency, their application remains limited to specific use cases. Nallakaruppan et al. (2024) [197] demonstrated how XAI can enhance the interpretability of RF models for predicting the Water Quality Index (WQI), which allows for more transparent and fair model outputs. However, these approaches have yet to be explored and adopted across several other hydrological domains.

The "black-box" nature of deep learning and ensemble-based ML models poses a major barrier to adoption in operational agencies, where decision-makers need to understand why a forecast is produced. Lack of interpretability may affect stakeholder confidence, particularly in high-stakes applications such as flood warnings or reservoir operations. Future advances must therefore prioritize explainability frameworks that make ML/DL forecasts interpretable, auditable, and defensible [198–200].

5.3. Model Generalization and Cross-Basin Transferability

Another major challenge concerns the limited ability of ML and DL models to generalize across regions. Because hydrological processes differ significantly between catchments, models trained on one basin often fail when applied to another. This reduces the usability of forecasting systems in ungauged or poorly monitored watersheds. Emerging methods such as transfer learning, domain adaptation, and physics-informed neural networks offer promising pathways to overcome this challenge and improve cross-basin forecast reliability [201–203].

Furthermore, the heterogeneity of watershed characteristics—soil properties, land cover, slope, hydrogeological conditions, and climate variability—creates structural dif-

ferences that ML/DL models struggle to capture. Even with large datasets, models often remain biased toward the hydrological signatures of their training basins [204,205]. A major research frontier is the development of globally trained hydrological models that can adapt to local conditions through minimal calibration. Advancing such systems requires standardized global datasets and improved representation of physical constraints within ML architectures, enabling models to learn both universal hydrological behaviors and region-specific nuances [201,206].

5.4. Non-Stationary Climate Conditions

Another challenge lies in forecasting under non-stationary climate conditions. As there has been frequent occurrences of intense hydrological events due to climate change, models trained on historical data may no longer be able to produce reliable projections [207,208]. To address this gap, several studies have utilized long-term forecasting models such as Neural Autoregressive Networks and ARIMA to capture the changing rainfall patterns in different regions, particularly in arid and semi-arid areas [188]. Du et al. (2025) [107] further emphasized that there is an ongoing need to adapt forecasting systems to account for the impacts of climate variability.

Additionally, climate non-stationarity disrupts traditional assumptions about seasonality, return periods, and hydrological extremes, which can lead to underestimation of flood risks or misrepresentation of drought persistence. Current forecasting systems often struggle to incorporate future climate scenarios, downscaled climate projections, or ensemble climate models in a seamless operational way [209–211]. Research must increasingly focus on developing climate-sensitive forecasting pipelines that dynamically update model parameters, integrate real-time climate anomalies, and use probabilistic forecasting to account for uncertainty. Such approaches will ensure that hydrological forecasts remain relevant as climatic patterns shift beyond historical baselines.

5.5. Computational Constraints and Operational Barriers

High-performing hybrid, deep learning, and ensemble models often require substantial computational resources, which limits their implementation in real-time forecasting settings. Many national and regional water agencies lack high-performance computing capacity, resulting in delays or operational inefficiencies [84,145]. Lightweight model architectures, cloud-based forecasting platforms, and automated data pipelines are needed to make advanced forecasting more accessible for routine operations.

In addition, the integration of high-resolution spatial data, ensemble weather predictions, and near-real-time data assimilation significantly increases computational load. These requirements often restrict the use of advanced models to research settings rather than operational environments [212–214]. Future developments should prioritize energy-efficient neural networks, scalable cloud-based platforms, and reduced-order hydrological models that retain predictive accuracy while minimizing computational expense. Investing in open-source, modular forecasting frameworks can also reduce dependence on expensive proprietary tools and increase accessibility across resource-limited agencies.

5.6. Multi-Source Data Integration Challenges

Modern forecasting relies on integrating data from remote sensing, IoT sensors, river gauges, weather radars, and reanalysis products. These datasets vary in spatial resolution, temporal frequency, and accuracy. Harmonizing multi-source data poses a substantial challenge, especially when dealing with missing values, sensor noise, and cross-platform inconsistencies. Future research must focus on real-time data assimilation frameworks and uncertainty quantification methods to improve model reliability [215–218].

Moreover, the increasing volume of hydrological big data generated by satellites, dense sensor networks, and national weather systems requires sophisticated data management architectures. Many agencies still rely on manual data cleaning or fragmented data storage, leading to delays and inconsistencies in operational forecasting [219,220]. Building unified hydrological data lakes, adopting standardized metadata conventions, and implementing automated quality-control algorithms will be essential for enabling seamless data integration. This is particularly important for hybrid forecasting systems that require synchronized spatial–temporal datasets to operate effectively.

5.7. Need for Standardized Evaluation and Benchmarking

Across the literature, studies employ different validation periods, performance metrics (NSE, RMSE, KGE, MAE, etc.), training procedures, and input structures. This lack of standardization limits meaningful comparison of model performance [221,222]. Developing shared benchmarking datasets, standardized testing protocols, and community-driven evaluation platforms would significantly improve quality and reproducibility in the field.

Furthermore, inconsistent evaluation strategies can lead to misleading interpretations of model performance, especially when models are tested only in ideal conditions or when validation datasets are too limited [223–225]. Establishing multi-basin benchmarking environments—similar to CAMELS datasets or HEPEX initiatives—will allow researchers to evaluate forecasting models under varied climatic, hydrological, and physiographic settings. Such benchmarking will not only promote transparency but also allow researchers to identify which approaches perform best for specific hydrological applications, ultimately guiding operational adoption.

Improving the quality and availability of data, expanding the uses of explainability frameworks, and creating models that account for non-stationary climate circumstances are all crucial steps in the development of hydrological forecasting systems. Forecasting models will continue to be reliable, flexible, and helpful for planning water resources and being ready for potential risks to these areas of focus.

5.8. Deepening Standardized Evaluation: Existing Frameworks, Gaps, and Actionable Pathways

Standardized evaluation has long been recognized as essential for meaningful comparison of hydrological forecasting models, yet its practical implementation remains uneven across methods, regions, and application contexts. While previous sections identify the lack of unified benchmarks as a key challenge, a deeper examination of existing evaluation frameworks reveals both progress and persistent limitations.

Several large-sample and community-driven frameworks have been developed to promote standardized evaluation. Dataset initiatives such as CAMELS and its regional extensions provide harmonized hydroclimatic inputs and benchmark basins, enabling cross-basin comparison of model performance under consistent conditions [226]. Community challenges and intercomparison projects further contribute by establishing shared datasets, predefined evaluation metrics, and transparent reporting protocols. Where applied, these frameworks have significantly improved reproducibility, facilitated objective comparison among statistical, physical, and data-driven models, and accelerated methodological development.

Despite these advances, the application effects of standardized evaluation remain uneven. Most existing frameworks are concentrated in data-rich regions, particularly North America and Europe, while data-scarce and climate-vulnerable regions—including parts of Africa, South America, and small island states—remain underrepresented [227]. As a result, models validated using standardized benchmarks may not generalize well to regions with sparse observations, complex terrain, or strong human-water interactions. Similarly,

standardized evaluation is often focused on streamflow forecasting, with limited extension to drought indices, water quality, or compound extreme events.

Scenario coverage also remains limited. Many benchmarks rely on historical data and stationary assumptions, offering insufficient evaluation of model robustness under climate change, land-use transitions, and extreme events. This constrains the ability of standardized metrics to inform model selection for long-term planning and climate adaptation.

To address these gaps, several actionable steps are recommended. First, standardized evaluation frameworks should be expanded geographically through the integration of satellite-based observations, reanalysis products, and regional data-sharing initiatives, enabling inclusion of underrepresented basins. Second, evaluation protocols should move beyond single-metric accuracy assessments to incorporate multi-criteria indicators, including robustness under non-stationarity, computational efficiency, interpretability, and uncertainty representation. Third, scenario-based benchmarking using climate projections, synthetic extremes, and stress-testing approaches should be systematically incorporated to evaluate model performance beyond historical conditions. Finally, open-access platforms and community-driven evaluation pipelines can lower barriers to adoption and promote consistent reporting across studies.

By shifting from problem identification to solution-oriented implementation, standardized evaluation can evolve from a conceptual ideal into a practical foundation for robust, transferable, and decision-relevant hydrological forecasting [228].

5.9. Synthesis of Challenges and Future Directions

The analysis of recent literature reveals that the advancement of hydrological forecasting is closely tied to addressing several interconnected technical and operational challenges. Table 8 synthesizes the key issues identified in Sections 5.1–5.7 and aligns them with corresponding future directions, providing a consolidated overview of the limitations that currently constrain forecasting performance and the pathways for enhancing model robustness, scalability, and operational value. The table highlights how persistent constraints such as data scarcity, limited model transparency, weak generalization across basins, climate non-stationarity, computational demands, multi-source data integration difficulties, and inconsistent evaluation standards collectively shape the effectiveness of hydrological prediction systems.

Together with the conceptual overview presented in Figure 2, which illustrates the broader role of forecasting tools in water resource management, Table 8 demonstrates that modern forecasting must evolve beyond single-model prediction toward an integrated, data-rich, and application-oriented framework. As hydrological extremes intensify under climate change, forecasting systems must not only provide accurate predictions but also support operational decision-making, early warning mechanisms, and long-term planning.

Looking forward, advances in monitoring infrastructure, explainable and physics-guided AI, global benchmark datasets, climate-adaptive modeling, computational efficiency, and standardized evaluation protocols will be essential for building forecasting systems that are reliable across regions and resilient to evolving hydroclimatic conditions. By focusing on these future directions, hydrological forecasting can continue to strengthen its role in disaster preparedness, water allocation, environmental protection, and sustainable water resources management.

Table 8. Summary of the major challenges identified across recent hydrological forecasting literature and the corresponding future research directions.

Challenge Category	Summary of Key Issues	Corresponding Future Directions
Section 5. Data Limitations and Infrastructure Constraints	Sparse hydrometeorological data, limited gauge networks, inconsistent calibration, missing records, lack of real-time systems; restricted adoption of advanced models in data-poor regions.	Improve monitoring networks; integrate IoT and remote sensing; build unified data repositories; implement automated quality control; expand long-term and high-resolution datasets.
Section 5.2. Limited Model Transparency and Trustworthiness	ML/DL models behave as "black boxes", limiting interpretability and operational trust; XAI only applied in narrow applications like WQI prediction.	Develop explainable AI methods; integrate physics-guided ML; improve model auditability; enhance stakeholder confidence; promote transparent operational frameworks.
Section 5.3. Model Generalization and Cross-Basin Transferability	ML/DL models overfit local hydrological conditions and fail when applied to ungauged basins; heterogeneous watershed characteristics hinder transferability.	Apply transfer learning and domain adaptation; create global benchmark datasets; develop physics-informed neural networks; explore minimal-calibration, globally trained models.
Section 5.4. Non-Stationary Climate Conditions	Climate change disrupts historical patterns, reducing reliability of models trained only on past data; increasing extremes and shifting regimes challenge forecast stability.	Use downscaled climate projections; implement adaptive/online learning; adopt probabilistic and scenario-based forecasting; integrate climate anomalies and ensemble climate models.
Section 5.5. Computational Constraints and Operational Barriers	Advanced hybrid and deep learning models require high computing power, limiting real-time use in agencies lacking HPC resources; heavy data assimilation increases processing time.	Develop lightweight model architectures; use cloud-based platforms; create reduced-order/efficient models; optimize data pipelines; promote open-source forecasting frameworks.
Section 5.6. Multi-Source Data Integration Challenges	Difficulties merging diverse data (IoT, satellite, radar, gauges) with different temporal and spatial resolutions; noisy or missing data reduce reliability.	Implement real-time data assimilation; standardize data formats; automate QC processing; build hydrological data lakes; incorporate uncertainty quantification for multi-source fusion.
Section 5.7. Need for Standardized Evaluation and Benchmarking	Inconsistent metrics, validation windows, and datasets reduce comparability across studies; limits reproducibility and objective model comparison.	Establish shared multi-basin benchmarks; unify evaluation metrics; use community-driven testing protocols; expand open-access datasets for model comparison and validation.

5.10. Integrated Discussion: Paired Challenges, Solution Pathways, and Regional Implementation

While Sections 5.1–5.7 discuss key challenges and future directions in hydrological forecasting, these issues are inherently interconnected and best understood through a paired challenge–solution perspective. To improve clarity and practical relevance without restructuring the preceding sections, Table 9 synthesizes the major challenges identified in this review, aligns them with targeted solution pathways, and highlights representative regional implementation contexts.

Table 9. Paired challenges, solution pathways, and regional implementation contexts in hydrological forecasting.

Key Challenge	Targeted Solution Pathway	Representative Regions
Data scarcity	Remote sensing, IoT, reanalysis integration	Africa, SE Asia, South America
Limited interpretability	Explainable AI, physics-guided ML	Europe, North America
Poor generalization	Transfer learning, global benchmarks	Asia, Global
Climate non-stationarity	Adaptive, probabilistic forecasting	Middle East, Australia
Computational constraints	Cloud-based, lightweight models	Developing regions
Data fusion complexity	Standardized data assimilation	Europe, East Asia
Inconsistent evaluation	Unified benchmarking platforms	Global

Data scarcity and monitoring limitations, discussed in Section 5.1, remain a dominant constraint in many developing and climatically vulnerable regions. Corresponding solution pathways emphasize multi-source data integration, including remote sensing, reanalysis products, and IoT-enabled monitoring systems. These approaches are already being operationalized in parts of Africa, Southeast Asia, and South America, where satellite-driven rainfall and streamflow estimation compensate for sparse gauge networks.

Challenges related to limited model transparency and stakeholder trust (Section 5.2) are increasingly addressed through explainable artificial intelligence and physics-guided learning frameworks. Such solutions are particularly relevant in regions with strong regulatory and institutional requirements, such as Europe and North America, where interpretable forecasts are essential for flood risk management, reservoir operations, and policy-driven decision-making.

Model generalization and transferability issues highlighted in Section 5.3 are paired with emerging solutions based on transfer learning, domain adaptation, and global benchmark datasets. Large-sample hydrology initiatives and continental-scale datasets have enabled scalable forecasting applications in Asia and other data-rich regions, while also supporting extrapolation to ungauged basins.

Climate non-stationarity (Section 5.4) necessitates adaptive and probabilistic forecasting strategies that incorporate climate projections and uncertainty-aware modeling. These approaches are increasingly applied in arid and semi-arid regions such as the Middle East and parts of Australia, where long-term water scarcity and extreme climate variability pose persistent risks.

Operational and computational constraints (Section 5.5) are addressed through cloud-based forecasting platforms, lightweight model architectures, and reduced-order modeling techniques. Such solutions enhance real-time forecasting capabilities in resource-limited agencies and rapidly urbanizing regions. Similarly, challenges in multi-source data fusion (Section 5.6) are mitigated through standardized data assimilation frameworks, particularly in regions with advanced radar and satellite infrastructure, such as East Asia and Europe.

Finally, the lack of standardized evaluation and benchmarking practices discussed in Section 5.7 is increasingly addressed through shared datasets, unified performance metrics, and community-driven evaluation platforms. These initiatives support objective model comparison and accelerate the transition from research to operational forecasting.

By explicitly linking challenges, solution pathways, and regional implementation contexts, this integrative discussion reinforces the practical implications of Section 5 and supports informed selection and deployment of hydrological forecasting frameworks across diverse environmental and institutional settings.

6. Summary and Conclusions

Hydrological forecasting has undergone a remarkable transformation in recent years, driven by the incorporation of cutting-edge technologies such as machine learning, deep learning, remote sensing, and hybrid physics–AI models. This evolution has not only enhanced the traditional statistical and physically based approaches but has also ushered in a new era of forecasting capabilities. The integration of these advanced methodologies has led to substantial improvements in the accuracy, timeliness, and adaptability of hydrological forecasting systems.

These sophisticated forecasting tools are now pivotal in a myriad of applications, including but not limited to flood monitoring and early warning systems, drought prediction, streamflow assessment, water quality monitoring, and strategic water allocation planning. For instance, in flood management, real-time data analysis enables authorities to issue timely alerts, potentially saving lives and minimizing property damage. Similarly, drought prediction models leverage historical data and climate patterns to help manage water resources more effectively, ensuring sustainability even in periods of scarcity.

However, despite these significant advancements, the field of hydrological forecasting still faces several critical challenges that impede the widespread operational adoption of these modern tools. One major hurdle is data limitations, as the availability and quality of hydrological data can vary greatly across different regions, impacting model performance. Additionally, the interpretability of complex models, particularly those based on deep learning, poses a challenge for practitioners who need to understand and trust the predictions being made.

Another pressing issue is climate non-stationarity, where changing climate conditions can render historical data less relevant for future predictions, complicating model training and validation. Furthermore, the applicability of models across different basins remains a concern, as hydrological processes can differ significantly based on local geography and climate. Finally, computational constraints, including the need for substantial processing power and resources, can limit the implementation of these advanced forecasting systems in real-time scenarios.

To fully harness the potential of modern hydrological forecasting tools, it is imperative to address these challenges head-on. Developing robust solutions that enhance data accessibility, improve model transparency, adapt to changing climatic conditions, and ensure cross-basin applicability will be essential. By overcoming these obstacles, we can build resilient and effective forecasting systems that support informed decision-making in the face of increasing hydrological and climatic uncertainty, ultimately contributing to better water resource management and disaster preparedness.

From this review, several key conclusions can be drawn:

1. Foundational models remain essential: Traditional statistical and physically based models provide critical insights into hydrological processes, but their standalone capabilities are limited in representing complex nonlinear interactions and rapidly changing climate-driven dynamics.
2. Machine learning and deep learning significantly enhance predictive performance by learning complex, nonlinear, and high-dimensional relationships from large hydrometeorological datasets. These models excel at capturing spatiotemporal patterns, improving forecast accuracy and lead time, and enabling scalable, data-driven forecasting across diverse hydrological applications. Their strengths are particularly evident in short- to medium-term forecasting and real-time applications, where rapid computation and adaptability to large data volumes are critical. However, their effectiveness remains strongly dependent on data availability, training representativeness, and model interpretability.

3. Hybrid and physics-informed models extend the capabilities of purely data-driven approaches by embedding hydrological process knowledge and physical constraints into learning frameworks. Rather than focusing solely on predictive accuracy, these models enhance robustness, physical consistency, and cross-basin generalization, especially in data-scarce or non-stationary environments. By coupling data-driven flexibility with process-based realism, hybrid approaches reduce overfitting, improve interpretability, and increase operational trust—making them particularly suitable for long-term planning, climate-sensitive forecasting, and decision-critical water management applications.
4. Regional differences matter: Model performance and applicability vary with local climate, topography, hydrological regime, and data availability. Region-specific calibration and validation are crucial to ensure reliable hydrological forecasts and effective decision support.
5. Persistent challenges remain: Data scarcity, limited transparency, and climate non-stationarity restrict model reliability, highlighting the need for expanded monitoring networks, explainable AI techniques, and climate-adaptive modeling frameworks.
6. Operational implementation considerations: To ensure real-world utility, forecasting systems must improve computational efficiency, seamlessly integrate multi-source data, and adopt standardized benchmarking for scalability and comparability in real-time applications.

In conclusion, the future of hydrological forecasting is poised to evolve significantly through the implementation of integrated, transparent, and climate-resilient systems. These systems will not only merge traditional hydrological knowledge with cutting-edge analytical tools but will also emphasize the importance of addressing existing limitations in current forecasting methodologies. By explicitly accounting for regional hydrological variations and the unique characteristics of different watersheds, AI-enhanced models can deliver nuanced and actionable insights.

Such advancements will play a crucial role in enhancing flood preparedness, enabling more effective drought management, and facilitating sustainable water allocation practices. Furthermore, these models will support long-term climate adaptation strategies, equipping stakeholders with the necessary information to make informed decisions regarding water resource management. Ultimately, this holistic approach will foster more resilient water management systems, ensuring that communities are better prepared to face the challenges posed by climate change and variability in water availability.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
ANFIS	Adaptive Neuro-Fuzzy Inference System (ANFIS)
ANN	Artificial Neural Network
ARIMA	Auto-regressive Integrated Moving Average
CIV	Variable Infiltration Capacity
CN	Curve Number
DL	Deep Learning
DT	Decision Trees
EFAS	European Flood Awareness System
GBM	Gradient Boosting Machine
GLM	Generalized Linear Model
GNN	Graph Neural Network
GT	Gamma Test
HEC-HMS	Hydrologic Engineering Center's Hydrologic Modeling System
IoT	Internet of Things
KNN	K-Nearest Neighbors
LSTM	Long Short-term Memory
MARS	Multivariate Adaptive Regression Splines
MCA	Multi-criteria analysis
MIKE-SHE	MIKE Système Hydrologique Européen
ML	Machine Learning
NAEFS	North American Ensemble Forecast System
NARX	Nonlinear Autoregressive with Exogenous Input
NH_3-N	Ammonia Nitrogen
NMME	North American Multimodel Ensemble
NSE	Nash-Sutcliffe Efficiency
NWP	Numerical Weather Prediction
RF	Random Forests
RNN	Recurrent Neural Network
RRKHS	Reproducing Kernel Hilbert Space
RVM	Relevance Vector Machine
SHAP	Shapley Additive Explanations
STA-GRU	Spatio-Temporal Attention Gated Recurrent Unit
SVM	Support Vector Machine
SWAT	Soil and Water Assessment Tool
WQI	Water Quality Index
XAI	Explainable Artificial Intelligence
XGBOOST	eXtreme Gradient Boosting

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